

ABHINAV JAGAN POLIMERA

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Education

Birla Institute of Science and Technology, Pilani - Hyderabad Campus

2020-2024

- B.E Computer Science - 8.2/10 CGPA

Publications

Remote Drone surveillance- Machine learning/ image processing

Publication Link

- **Researched and implemented** an automated surveillance system using **Python, YOLO** to detect and classify objects in real time; designed **threshold-based algorithms** using **object count** and **activity duration** to automatically flag suspicious actions, with **95% accuracy** further strengthening remote security threat detection.

Performance Assessment of Neural Radiance Fields and Photogrammetry

Publication Link

- **Conducted** a comparative study between **Neural Radiance Fields(99.89% 3D reconstruction)** and traditional **Photogrammetry(99% 3D reconstruction)** of man-made and natural features, leveraging **neural networks** and **volumetric rendering**; the work was selected and presented at the **ICAART** conference.

Work Experience

Cisco Systems

Apr 2026 - Present

Software Developer II — AI Infrastructure & Distributed Systems

On-site

- Engineered to an **AI-driven Infrastructure-as-a-Service** initiative for deploying **quantized ML models** which are of **0.5-2MB** and have a **CPU utilization under 2%** on **Cisco routers**; applied **data-driven model selection** on large-scale network telemetry to identify patterns and choose optimal ML approaches; **delivered concept-to-prototype with a 4-member core team**; **recognized across Cisco AI forums** and currently in customer evaluation phase for deployment.
- Architected and developed a **semantic knowledge graph** for the **Cisco Silicon One router operating system**, converting the codebase into a **structured, agent-readable knowledge graph**. Optimized **LLM context retrieval precision by ~30%** and **token efficiency**, enabling AI agents to reason over the codebase with higher accuracy and produce more reliable results in **agentic software development workflows**.
- **Spearheaded** the deployment of a scalable automation infrastructure using open-source and Cisco internal frameworks. Built **AI-powered developer platforms**, including a **metrics analytics dashboard (100+ users)** and a unified **traffic mirroring platform with O(1) support-matrix validation, drag-and-drop router simulation, and MCP-based modular components**, reducing issue reproduction time by **80% (5 → 1 day)**.
- Led the **end-to-end design and delivery of two customer-driven traffic mirroring features** for **Cisco 8000 (Silicon One) routers**: (1) **egress traffic mirroring** within the router network for **SoftBank**, and (2) a **next-generation ERSpan** solution introducing **UDP encapsulation** as a scalable alternative to **GRE** for **Microsoft**, reducing tunnel overhead by **16%**, saving **~5 PB/year** of telemetry traffic, improving monitoring-fabric capacity by **2.3%**, and leveraging **AI-assisted development** to accelerate delivery by **60%**.
- **Optimized router OS feature testing and deployment** through design refinement and **parallel execution** leading to **script execution time to drop by 90%**; the initiative earned an internal **Cisco excellence award**.

Projects

Playlistify - web development

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- Identified a product usability gap in Spotify and independently built Playlistify, a **Python- and Flask-based web app** leveraging the **Spotify Web API**, that converts liked songs into shareable playlists with a single click, demonstrating strong problem-solving initiative, curiosity, and full-stack development skills.

Augmented Reality Assisted Lego Construction

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- Built an **AR LEGO assembly assistant** with **real-time guidance**; curated a dataset and trained a **state-recognition model** for partial builds using **object detection** and **computer graphics**.

Skills

Programming: Python, C, C++, Shell Scripting

AI & Machine Learning: Large Language Models (LLMs), Agentic AI, AI Agents, Multi-Agent Systems, Agentic Workflows, Context-Prompt Engineering, Retrieval-Augmented Generation (RAG), Knowledge Graphs, Vector Databases, Embedding Models, Fine-tuning, LLM Evaluation (Evals), Model Context Protocol (MCP), LangGraph, Hugging Face, Transformers

Systems & Networking: Distributed Systems, Network Operating Systems, Systems Programming, Computer Networking, Routing, Cisco Silicon One, Linuxg

Computer Vision: OpenCV, Computer Vision, Image Processing, Object Detection, Neural Radiance Fields (NeRFs)